

Retrofitting aging commercial gas boilers with a high-efficiency hydronic split

Comprehensive Hardware Design Analysis

Revision 1 | Generated 04/05/2026

Executive Summary

Generated

Project ID	a_compact_commercial_airsource_heat_pump
Industry Domain	commercial_hvac
Engine Version	PDF Engine v2.1 (Full Detail)
Target Quantity	Not specified

ARCHITECTURE AT A GLANCE

MODULES 0	BILL OF MATERIALS ROWS 0	IDENTIFIED RISKS 0
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ECONOMICS

ESTIMATED UNIT COST £0.00	TARGET CEILING £4,500.00	HEADROOM £4,500.00
TOTAL NON-RECURRING ENGINEERING £0.00	UNIT COST PLUS AMORTISED NON-RECURRING ENGINEERING £0.00	

Cover

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1. Brief & Requirements

1.1 Overview & Context

Subject / Use Case

Retrofitting aging commercial gas boilers with a high-efficiency hydronic split heat pump to provide sustainable space heating and domestic hot water for small commercial properties.

Mission Statement

To deliver an affordable, highly efficient, and future-proof R290 air-source heat pump that accelerates the decarbonization of UK small commercial buildings.

Target Customers

Small-to-medium enterprise (SME) facility managers, HVAC installers, MEP consultants, commercial landlords, and local government retrofit programs.

Why Now (Market Timing)

Tightening UK F-Gas regulations are forcing a shift away from high-GWP refrigerants, while Minimum Energy Efficiency Standard (MEES) mandates are forcing commercial landlords to immediately upgrade their heating systems.

Full Synthesis Report

The UK market for commercial air-source heat pumps (ASHPs) is in a phase of accelerated growth, mandated by the government's net-zero targets and the imminent phase-out of fossil-fuel boilers in commercial real estate. Developing a 30kW thermal output system specifically engineered for small commercial buildings provides an ideal solution for retrofitting offices, retail spaces, and light industrial units. The strategic utilization of R290 (propane) as the primary refrigerant is a key differentiator; its Ultra-Low Global Warming Potential (GWP of 3) future-proofs the product against stringent UK and European F-Gas regulations that are rapidly phasing out HFCs like R32 and R410A. Achieving a COP of ≥ 3.5 at A2/W35 is highly feasible with R290, which possesses superior thermodynamic properties, especially for producing reliable flow temperatures without aggressive compressor staging. However, utilizing a hydronic split architecture with R290 introduces specific design and safety challenges. Routing highly flammable, A3-class propane requires rigorous

Data Sources:

[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

adherence to BS EN 378 and DSEAR regulations. To mitigate this safely in a split design, the primary sealed refrigeration circuit must remain predominantly situated in the outdoor unit, utilizing a safe hydronic transfer to the indoor unit, or implementing ATEX-rated enclosures and active ventilation. Achieving the target unit cost of £4,500 at a volume of 500 units per year is aggressive but viable through modular lean assembly, sourcing off-the-shelf inverter-driven compressors, and a localized European/UK supply chain. Achieving MCS (Microgeneration Certification Scheme) certification is absolutely critical; it ensures quality compliance and unlocks access to crucial government retrofit grants and commercial tax incentives, making the economic proposition undeniable for commercial landlords.

Data Sources:

[User] Founder brief text

[LLM] Gemini 3.1 Pro — research synthesis

1.2 Market & Competitors

Competitor Landscape

Vaillant

Product	aroTHERM plus (Cascade)
Pricing Strategy	£8,500+ (for cascade setup)
Technical Specs	Multiple unit cascade to reach 30kW, R290 refrigerant, COP 3.6 at A2/W35.

STRENGTHS

Exceptional build quality, established brand, exceptionally quiet operation.

WEAKNESSES

High capital cost for 30kW, relies on complex cascading of residential units rather than one commercial unit.

DIFFERENTIATION ANGLE

Providing a native, single-unit 30kW thermal output at nearly half the total installation cost.

Clivet

Product	Edge EVO 2.0 EXC
Pricing Strategy	£6,500
Technical Specs	30kW standalone, R32 refrigerant, COP 3.4 at A2/W35.

STRENGTHS

Good footprint, integrated hydronic module, commercial focus.

WEAKNESSES

Uses R32 which has higher GWP and faces phase-down, slightly lower A2/W35 efficiency.

DIFFERENTIATION ANGLE

Future-proof R290 refrigerant and superior SCOP value.

Data Sources:

[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

Mitsubishi Electric

Product	Ecodan CAHV / QAHV
Pricing Strategy	£12,000+
Technical Specs	40kW heating, CO2 or R407C refrigerant, high temp up to 90°C.

STRENGTHS

High capacity, incredibly reliable, capable of very high DHW temperatures.

WEAKNESSES

Extremely expensive, optimized for large volume hot water rather than low-temp space heating.

DIFFERENTIATION ANGLE

Aggressively priced at £4,500 specifically for standard space heating retrofits where 90°C is unnecessary.

Data Sources:

[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

1.3 Engineering Constraints

Core Targets

Unit Cost Ceiling	£4,500.00
Max Mass	-
Target Process	
Target Material	
Tolerance Target	
Production Quantity	
Batch Size	500

Research Sources

TITLE	TYPE	YEAR	RELEVANCE
MCS MIS 3005-I Heat Pump Standard	Regulatory Standard	2026	Outlines the strict installation and product performance requirements vital for obtaining UK government grants.
UK F-Gas Regulation Guidelines	Government Document	2026	Provides legal context for the phase-down of HFCs and validates the strategic choice of ultra-low GWP R290.
BS EN 378: Refrigerating systems and heat pumps	Technical Standard	2026	Mandates the safety and environmental requirements for designing systems with highly flammable A3 refrigerants like propane.
BSRIA Heat Pump Market Report 2023	Market Report	2026	Provides detailed UK commercial market sizing, competitor pricing, and adoption rates for 30kW-range commercial heat pumps.
CIBSE AM17: Heat pumps in large non-domestic buildings	Design Guide	2026	Offers best practices and system architecture guidelines for specifying heat pumps in commercial applications.

Data Sources:

[User] Founder brief text

[LLM] Gemini 3.1 Pro — research synthesis

2.1 MCS 007

Microgeneration Certification Scheme - Product Certification

Status: not-started

Owner: Compliance Engineer

Standard Summary

Product quality and performance validation for heat pumps sold in the UK.

Applicability Assessment

UK-wide for commercial/residential subsidy eligibility.

Engineering Design Impact

Requires verified EN 14511 testing for COP and acoustic emissions.

EVIDENCE REQUIRED

UKAS accredited lab test reports for COP and noise.

GAP ACTION

Engage a certified test lab to baseline a prototype against A2/W35 performance needs.

Data Sources:
[LLM] Gemini 3.1 Pro — standards extraction

2.2 BS EN 378

Refrigerating systems and heat pumps safety

Status: not-started

Owner: Mechanical Engineer

Standard Summary

Mandates safety, environmental, and design factors for refrigeration equipment.

Applicability Assessment

Specifically impacts the split architecture and indoor unit due to R290 (A3 flammable) usage.

Engineering Design Impact

Requires ATEX mitigation, leak detection sensors, and potentially isolating the R290 completely outside (hydro-split).

EVIDENCE REQUIRED

Explosion risk assessment, fail-safe leak detection documentation.

GAP ACTION

Conduct an A3 flammability risk assessment for the proposed indoor-outdoor interconnects.

Data Sources:

[LLM] Gemini 3.1 Pro — standards extraction

2.3 UK SI 2016 No. 1101

Electrical Equipment (Safety) Regulations 2016

Status: not-started

Owner: Electrical Engineer

Standard Summary

UK equivalent of the LVD, ensuring electrical safety of connected components.

Applicability Assessment

All electrical/electronic components within both indoor and outdoor units.

Engineering Design Impact

Requires specific grounding, insulation schemes, and short-circuit protection for the inverter module.

EVIDENCE REQUIRED

Complete Technical File and UKCA mark test reports.

GAP ACTION

Review inverter and compressor data sheets for UK compliance.

Data Sources:

[LLM] Gemini 3.1 Pro — standards extraction

2.4 UK SI 2010 No. 2617

Ecodesign for Energy-Related Products

Status: not-started

Owner: Systems Engineer

Standard Summary

Minimum energy efficiency requirements (ErP) for space heaters.

Applicability Assessment

Mandatory for commercial sale in the UK.

Engineering Design Impact

System must achieve stringent Seasonal Space Heating Energy Efficiency (SCOP) thresholds.

EVIDENCE REQUIRED

ErP label data, seasonal COP calculation and laboratory verification.

GAP ACTION

Perform preliminary SCOP modeling on the compressor selected.

Data Sources:

[LLM] Gemini 3.1 Pro — standards extraction

2.5 DSEAR 2002

Dangerous Substances and Explosive Atmospheres Regulations

Status: not-started

Owner: Regulatory Lead

Standard Summary

Protects against risks from fire, explosions and similar events arising from dangerous substances in the workplace.

Applicability Assessment

Applies to the manufacturing facility and commercial deployment sites using R290.

Engineering Design Impact

Dictates safe ventilation zoning around the outdoor unit and indoor pipework.

EVIDENCE REQUIRED

DSEAR risk assessment and Zone classification drawing.

GAP ACTION

Draft installation manuals with precise exclusion zone dimensions for installers.

Data Sources:

[LLM] Gemini 3.1 Pro — standards extraction

3. Sizing & Spatial Optimisation

Sizing constraints and allocations not generated.

4. System Modules & Architecture

No modules defined in the system architecture.

5. Bill of Materials

No parts in bill of materials.

6. Cost Waterfall & Economics

UNIT COST £0.00	TARGET CEILING £4,500.00	HEADROOM £4,500.00
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Make vs Buy Split

PURCHASED (BUY) TOTAL £0.00	CUSTOM (MAKE) TOTAL £0.00	MAKE % 0%
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Per-Module Roll-up

Per-module cost data not available.

Non-recurring engineering and overheads

Total non-recurring engineering (Tooling, £0.00
Safety Testing, Compliance)

Grand Total (Unit + Non-recurring Engineering)	£0.00
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Data Sources:
[Deterministic] Domain overhead multiplier model

Cost Confidence Assessment

Costing methodology employs deterministic expansions over a historical parts library combined with spot-price indices for raw materials. Custom parts are calculated volumetrically and adjusted by domain-specific complexity multipliers to ensure realistic unit economics.

Make-versus-buy categorisation is automated based on process availability. Non-recurring engineering (NRE) totals reflect tooling, testing, and compliance costs amortised over the target batch size. Margins of error apply to COTS components subject to global supply chain volatility.

Data Sources:

[Deterministic] Domain overhead multiplier model

7. Risks Register

No risk entries declared in the system architecture.

8. Supplier Shortlist

No supplier matches found during the search phase.

9. Audit Log

Pipeline execution log — generated by PDF Engine v2.1.

This document was generated automatically. Contents represent the synthesised output of the specialist pipeline.

Pipeline Execution Summary

Project Identifier	a_compact_commercial_airsource_heat_pump
Research Result	Generated
System Modules	0 Modules
Bill of Materials Lines	0 Parts
Cost Analysis	Pending
Specialist Reviews	0 Completed
Supplier Shortlists	0 Matched

10. Source Attribution

This table documents the origin of data for each major section of the report, ensuring traceability of LLM outputs versus deterministic or user-provided data.

SECTION	SOURCE	PROVIDER / SYSTEM	DETAIL
Research	LLM	OpenRouter	Gemini 3.1 Pro via OpenRouter
Research	USER	user	Founder brief text
Regulatory	LLM	Unknown LLM	Gemini 3.1 Pro — standards extraction